Graph Theory & Networks (AI HL)

Section 1: Degrees and the Handshaking Lemma

1 What does the Handshaking Lemma state for any undirected graph with V vertices and E edges?

a. The sum of the degrees of all vertices equals twice the number of edges: $\sum \deg(v) = 2E$

b. The sum of the degrees equals E

c. The number of vertices with odd degree is always odd

d. Every vertex must have degree at least 2

2 A graph has 6 vertices, each having degree 3. How many edges does the graph have?

a. 9

b. 18

c. 6

d. 12

3 Why can a graph never have an odd number of vertices of odd degree?

a. Because $\sum \deg(v) = 2E$ is always even, the sum of odd degrees must be even, forcing an even count of odd vertices

b. Because edges cannot cross each other

c. By Euler's formula $V - E + F = 2$

d. Because every vertex must have at least one adjacent edge

Section 2: Eulerian and Hamiltonian Paths and Circuits

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4 What condition is both necessary and sufficient for a connected graph to contain an Eulerian circuit (a closed trail visiting every edge exactly once)?

a. Every vertex has an even degree

b. Exactly two vertices have odd degree

c. The graph contains no cycles

d. The graph is a complete bipartite graph

5 If a connected graph has exactly two vertices of odd degree, what type of Eulerian walk does it contain?

a. An Eulerian trail (open path visiting every edge once) that starts at one odd vertex and ends at the other

b. An Eulerian circuit returning to the start

c. No Eulerian trail or circuit can exist

d. A Hamiltonian cycle

6 What is the difference between an Eulerian circuit and a Hamiltonian cycle?

a. An Eulerian circuit visits every edge once; a Hamiltonian cycle visits every vertex once

b. An Eulerian circuit visits every vertex once; a Hamiltonian cycle visits every edge once

c. They are identical concepts in graph theory

d. Hamiltonian cycles apply only to directed trees

Section 3: Minimum Spanning Trees (Kruskal & Prim)

7 In a connected weighted graph with V vertices, how many edges does any spanning tree contain?

a. $V - 1$

b. V

c. $V + 1$

d. $\frac{V(V-1)}{2}$

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8 How does Kruskal's algorithm select edges to construct a Minimum Spanning Tree (MST)?

a. Sorts all edges by weight and greedily adds the smallest available edge that does not create a cycle

b. Grows a single tree by continuously adding the closest vertex to the current tree

c. Removes the largest cycle edges one by one until only a tree remains

d. Visits vertices in alphabetical order

9 How does Prim's algorithm differ conceptually from Kruskal's algorithm?

a. Prim grows a single connected tree from an arbitrary starting vertex, whereas Kruskal maintains a forest of components until unified

b. Prim works only on unweighted graphs

c. Kruskal finds the longest path rather than minimum tree

d. Prim can only be applied to planar graphs

Section 4: Shortest Paths and Dijkstra's Algorithm

10 What is the purpose of Dijkstra's algorithm?

a. To find the shortest paths from a single source vertex to all other vertices in a network with non-negative edge weights

b. To find a minimum spanning tree

c. To determine if a graph has an Eulerian circuit

d. To find the maximum flow through a network

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11 Why does Dijkstra's algorithm fail if a network contains negative edge weights?

a. A greedy choice once finalized could be improved later by a negative edge, invalidating the optimality assumption

b. Distances cannot be negative by definition

c. The algorithm enters an infinite cycle immediately

d. Square roots of negative weights cannot be computed

12 In a planar graph, if $V = 6$ and $E = 9$, how many faces F does the graph enclose according to Euler's formula $V - E + F = 2$?

a. 5

b. 3

c. 4

d. 6

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